

TELCONG NETPLAN

Network Planning & Regulatory Affairs

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For incorporation into Section D, Spectrum band rationale narrative

ITU-R WP 5D Note on 3.5 GHz Harmonisation

Reading notes for the 5G CBD Pilot Section D, Spectrum band rationale

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Subject	ITU-R Working Party 5D position on 3300-3800 MHz harmonisation for IMT-2020
Source	ITU-R Report M.2516, Recommendation ITU-R M.1036-7
Use	Hearing agenda item 1, Spectrum allocation rationale

Background, the 3.5 GHz band globally

The 3300-3800 MHz frequency range, commonly called the 3.5 GHz band, has emerged as the global mid-band spectrum of choice for 5G NR deployment. The band offers a balance between coverage (the lower frequencies propagate more favourably than the millimetre-wave 5G bands) and capacity (the bandwidth available is significantly greater than the sub-1 GHz bands historically used for 4G coverage). At WRC-19, the International Telecommunication Union's World Radiocommunication Conference confirmed the 3300-3800 MHz range as a primary candidate for IMT-2020 (5G) deployment globally, although the exact sub-range available for IMT varies by region.

ITU-R Working Party 5D position

ITU-R Working Party 5D, which is responsible for the IMT terrestrial component, has issued a series of documents on 3.5 GHz harmonisation since WRC-19. The key documents are: Report ITU-R M.2516 (band characteristics for IMT in 3300-3800 MHz), Recommendation ITU-R M.1036-7 (frequency arrangements for IMT), and the ongoing studies in the WRC-27 preparatory cycle (which is considering further extensions in adjacent bands).

The WP 5D position can be summarised in five points:

- (1) The 3300-3800 MHz range is the primary 5G mid-band globally. Administrations are encouraged to make the maximum portion of this range available for IMT consistent with the protection of incumbent services.
- (2) Within the range, the 3400-3800 MHz sub-range is preferred for IMT because the 3300-3400 MHz sub-range has heavier incumbent radiolocation use in many countries.
- (3) For terrestrial IMT, the band is recommended for TDD deployment with a synchronisation framework to minimise cross-operator interference. The synchronisation can be either inter-operator (preferred) or via guard bands at operator boundaries.

(4) Adjacent-band coordination with satellite earth stations (FSS) in the 3700-4200 MHz range is required. WP 5D Recommendation M.2117 specifies the technical coordination criteria.

(5) The 3500-3600 MHz sub-range (referred to as the "lower portion" in Section D of our submission) is the cleanest portion within 3400-3800 MHz in most administrations, because it is furthest from FSS earth stations in 3700+ and from radiolocation in 3300-3400. WP 5D recommends this sub-range for early IMT deployment.

Nigerian deployment status

Nigeria has deployed 5G in the 3400-3600 MHz sub-range since 2022 under licences held by MTN Nigeria and Mafab Communications, won in the December 2021 spectrum auction. The 3600-3700 MHz sub-range remains available for new assignments and is the focus of our application. The 3700-3800 MHz upper portion is the subject of the 2025 9mobile pilot (see hearing transcripts archive, T3).

Our application requests 100 MHz contiguous in the 3500-3600 MHz lower portion. This sub-range is currently lightly used in Abuja FCT (MTN holds 80 MHz here for Lagos and Port Harcourt deployment, but has not extended into Abuja). The Commission has indicated in informal consultations that 3500-3600 MHz in Abuja FCT is available for assignment.

Adjacent band considerations

The application requires coordination with two adjacent assignments:

Adjacent band	Service	Holder / status	Coordination required?
3400-3500 MHz	IMT (5G NR)	MTN Nigeria (Lagos, Port Harcourt), not Abuja	No (different geographic area)
3600-3700 MHz	Unassigned in Abuja FCT	Available for future assignment	Future-looking
3700-3800 MHz	IMT (5G NR pilot)	9mobile pilot in Lagos, not Abuja	No (different geographic area)
3800-4200 MHz	Fixed Satellite Service (FSS) earth stations	Multiple licensees	Yes (Recommendation M.2117 criteria apply)
3300-3400 MHz	Radiolocation (legacy MoD)	Reserved for federal government use	Yes (Annex C-3 coordination filing)

The two coordination filings (with FSS licensees in 3800-4200 MHz, and with the Ministry of Defence for the 3300-3400 MHz radiolocation reservation) are at advanced stages. Documentation is at Annex C-3 and Annex C-4 of the submission. The FSS coordination has reached written agreement with three of the four affected earth station operators; the fourth is in negotiation and expected to conclude before the 60-day determination period ends.

Hearing position

Our hearing position on agenda item 1 (Spectrum allocation rationale) should anchor on the following narrative arc:

(a) The 3.5 GHz band is the global IMT mid-band per WRC-19 and ITU-R WP 5D guidance.

- (b) The 3500-3600 MHz lower portion is the recommended sub-range for early IMT deployment, with the least incumbent friction.
- (c) 3500-3600 MHz in Abuja FCT is currently unassigned and available.
- (d) The 100 MHz contiguous block is the standard 5G NR carrier size for mid-band deployment and matches the bandwidth used by other Nigerian operators in their 3.5 GHz deployments.
- (e) Coordination filings are in place with all affected incumbents and adjacent assignees.

END OF NOTES

For incorporation into Form NCC-7B Section D and into the hearing Q&A brief for agenda item 1.